Event Density: Empirical Predictions from a Substrate-Level Identification of $a_0 = cH_0/(2\pi)$

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May 2026

Event Density: Empirical Predictions from a Substrate-Level Identification of $a_0 = cH_0/(2\pi)$

Series: Wave-3 Empirical-Predictions Bundle (cross-arc; primary outreach paper) **Status:** Compilation of ten ED empirical predictions across gravity, GW, matter-wave interferometry, soft-matter, and analogue-gravity arcs. Three published Wave-3 prediction papers + seven session-ready experimental protocols + one peer-reviewed passed test + two postdictions + cross-scale falsifiers. Verdict across the bundle: every prediction is form-IDENTIFIED or form-FORCED from the substrate primitives; numerical content where present is INHERITED unless otherwise noted; falsification criteria explicit per prediction. **Date:** 2026-05-17 **Anchors:** Paper_087 (13 primitives); Paper_095 (form-FORCED / value-INHERITED methodology); Paper_027 (Newton's G); Paper_029 ($a_0 = cH_0/(2\pi)$); Paper_030 (ED Combination Rule); Paper_031 (BTFR slope-4); Paper_034 (deep-MOND asymptotic); Paper_036 (MOND field equation); Paper_038_5 (-smallness reframing; M3 retroactive); Paper_038_6 (Pred_WeakLensing_Activity); Paper_047 (Hawking spectrum); Paper_052_5 (Pred_MergerLag); Paper_089 (V1 retarded kernel); Paper_090 (V5 cross-chain kernel); Paper_ED_Cos_05 (Dark Energy; M3 unconditional); Paper_ED_GW_00 (GW); Universal Mobility Law evidence paper (peer-reviewed, May 2026); ED-SC 4.x arc; protocols in `predictions/`: QC-Mass-Extrapolation, FRAP-High-BSA, AFM-Dewetting-ED-SC, ED-Acoustic-BEC-Extremal, ED-Acoustic-EIT-Extremal, ED-Acoustic-OpticalPulse, ED-09.5 Participation-Envelope, Triad-Coupling-C7, ED-RLC-Analogue.

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§1 Introduction

The Baryonic Tully-Fisher Relation (BTFR) is one of the tightest empirical relations in galactic astrophysics: across roughly four decades of baryonic mass, the asymptotic flat rotation velocity v_{flat} of a galaxy satisfies $v_{\text{flat}}^n \propto M_b$ with $n = 3.95 \pm 0.08$ — consistent with $n = 4$ exactly, with scatter dominated by observational uncertainty. The Event Density (ED) substrate-ontology research program derives **BTFR slope exactly 4 with zero intrinsic scatter in the deep-MOND asymptote** from substrate primitives, with no halo modeling and no free parameters, via composition of Newton's G from substrate constants (Paper_027), the substrate-level identification of the MOND transition acceleration $a_0 = cH_0/(2\pi)$ (Paper_029), the ED Combination Rule $a = \sqrt{a_N \cdot a_0}$ (Paper_030, conditional on a single declared postulate P14), and the standard circular-orbit condition $v^2/R = a$.

The BTFR result is the wedge for this paper. The same substrate-level identification — that a_0 is the azimuthal-Fourier projection of the cosmic decoupling surface $R_H = c/H_0$ onto an accelerating chain's dipole mode, producing the geometric $1/(2\pi)$ structurally — generates a family of related predictions across the corpus's empirical-prediction sector. This paper bundles ten such predictions: three published Wave-3 prediction papers, seven session-ready experimental protocols, plus three postdictions (one peer-reviewed) and one cross-scale-covariation prediction. Each prediction is presented with its substrate-level mechanism, its sharpest falsifier, and (where applicable) the experimental platform and protocol status.

ED's methodological discipline (Paper_095) is *form-FORCED / value-INHERITED*: structural forms are derived from substrate primitives; numerical magnitudes are inherited from empirical matching. This paper inherits that discipline throughout. Where a prediction's numerical content is INHERITED rather than substrate-derived, it is named as such — not hidden.

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§2 BTFR slope-4 with zero intrinsic scatter (sharpest empirically-clean falsifier)

Prediction. $v_{\text{flat}}^4 = GM_b a_0$ exactly, with zero intrinsic scatter, in the deep-MOND asymptotic regime ($a_N \ll a_0$).

Substrate derivation chain (Paper_031):

1. Newton's law $a_N = GM/R^2$ with $G = c^3 \ell_P^2/\hbar$ (Paper_027).
2. Transition acceleration $a_0 = cH_0/(2\pi)$ (Paper_029; see §3 below for derivation).
3. ED Combination Rule $a = \sqrt{a_N \cdot a_0}$ in the joint weak-gradient regime (Paper_030, conditional on P14 Bilocal Strain Coupling).
4. Circular-orbit condition $v^2/R = a$ (standard kinematics).

Composition: $v^2/R = \sqrt{(GM_b/R^2) \cdot a_0} = (1/R)\sqrt{GM_b a_0}$, giving $v^2 = \sqrt{GM_b a_0}$ — radial dependence cancels exactly. Squaring: $v^4 = GM_b a_0$. **Slope exactly 4; intrinsic scatter exactly zero in the asymptote.**

Why this is sharp. Three competing frameworks attempt BTFR:

- **Dark-matter cosmology:** slope-4 emerges from coincidence in tuned halo profile distributions; tightness is a tuning coincidence.
- **MOND as phenomenology:** slope-4 is built in by construction.
- **ED substrate:** slope-4 is *derived* from a substrate-level identification (Paper_027 + Paper_029 + Paper_030 + circular-orbit kinematics); no halo modeling, no free parameters, no tuning.

Test. SPARC galaxy-rotation-curve catalog (Lelli, McGaugh, Schombert 2016): publicly available, well-characterized observational systematics. Slope-4 + zero-scatter prediction is directly testable against the deep-MOND asymptotic subset of SPARC. The McGaugh-Lelli-Schombert 2016 measurement $n = 3.95 \pm 0.08$ is *consistent with* slope-4; the residual scatter is *consistent with* observational uncertainty. ED predicts that any future increase in measurement precision tightens the slope toward 4 exactly and the residual scatter toward zero.

Sharpest falsifier (F1). Deep-MOND-asymptotic BTFR slope significantly different from 4 beyond observational uncertainty refutes Paper_030's ECR via P14, refutes Paper_031's composition, or refutes Paper_029's substrate-level identification of a_0. Equivalently: detection of intrinsic (non-observational) scatter in the deep-MOND-asymptotic BTFR refutes the radial-cancellation derivation. Currently consistent at all SPARC precision.

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§3 $a_0 = cH_0/(2\pi)$: the transition acceleration from substrate-level dipole projection

Prediction. The MOND-class transition acceleration is

$$a_0 = \frac{cH_0}{2\pi} \approx 1.08 \times 10^{-10} \text{ m/s}^2$$

with $H_0 \approx 70$ km/s/Mpc inherited from cosmology. The empirically measured MOND value is $a_0^{\text{MOND}} \approx 1.2 \times 10^{-10}$ m/s² — match within ~10% with **zero free parameters**.

Substrate derivation chain (Paper_029):

1. **Cosmic decoupling surface** at $R_H = c/H_0$ (Paper_028): beyond this radius, substrate-level cross-locus influence cannot reach a local chain coherently. This is the substrate-level effective boundary set by the kernel-propagation-rate / Hubble-expansion-rate crossover.
2. **Dipole-mode projection** of cosmic content onto an accelerating chain. An accelerating chain breaks the full substrate $SO(3)$ rotational symmetry to the residual $SO(2)$ symmetry about the acceleration axis. The leading anisotropic mode is the dipole ($l = 1$); higher multipoles are higher-order in $|\vec{a}|/c$.
3. **Azimuthal-Fourier normalization** of the dipole's $|m| = 1$ projection on the residual $SO(2)$ (Paper_029 §5.1 dipole-projection integral): the Fourier-mode normalization on the circle of period 2π produces the geometric factor $1/(2\pi)$.

Composition: $a_0 = cH_0 \cdot (1/(2\pi)) = cH_0/(2\pi)$. The factor of 2π is **geometrically forced** by azimuthal-Fourier-mode normalization, not phenomenologically fit.

Why this matters. Neither dark-matter cosmology nor MOND-as-phenomenology derives a_0. The empirical observation that $a_0 \approx cH_0$ has been recognized for decades as suggestive of a cosmological connection (Milgrom 1983 and subsequent), but no structural mechanism has been provided. ED supplies the mechanism via dipole projection of R_H onto an accelerating chain's anisotropic adjacency — and forces the precise $1/(2\pi)$ factor.

Sharpest falsifier (F2). Independent observational refinement of H_0 inconsistent with the substrate-derived $a_0 = cH_0/(2\pi)$ relation at observed MOND a_0 refutes Paper_029's dipole-projection mechanism. Hubble-tension-class shifts of H_0 should produce *synchronized* covariation of a_0 at the substrate-derived rate (linear in H_0). Hubble tension shifting H_0 without corresponding shift in observed a_0 would refute the substrate identification.

Post-Route-A status (2026-05-17). The Route A audit cascade closes H_0 at substrate-parameter-INHERITED level (Paper_027-analog) via $H_0 = (8\pi/3)^{1/2}\ell_P c^{1/2}\Theta_{\text{ED}}^{1/2}$. a_0 correspondingly tightens to: $a_0 = (c/(2\pi)) \cdot (8\pi/3)^{1/2}\ell_P c^{1/2}\Theta_{\text{ED}}^{1/2}$ — all factors substrate-parameter-INHERITED at primitive level (analog of Paper_027's inheritance of $\ell_P, \hbar, c$).

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§4 Weak-lensing activity-dependence (Paper_038_6)

Prediction. Weak-lensing convergence signals are **activity-dependent**: galaxies and clusters in high-activity states (star-forming, AGN-active, dynamically unrelaxed) produce enhanced weak-lensing signals compared to low-activity (quiescent, relaxed) systems matched in baryonic mass + total mass + redshift.

Substrate mechanism (Paper_038_6). High activity corresponds to large local V5 cross-chain mobility gradients ($\nabla\mu$). Per Paper_EDSC_AcousticMetric_CurvatureConcentration, acoustic-metric curvature scales as $R_{\rm ac} \propto |\nabla\mu|^2/\mu^2$. This modifies the deep-MOND interpolation function:

$$\mu_{\rm eff}(x) = \mu_{\rm MOND}(|\nabla\Phi|/a_0) \cdot [1 + \alpha_{\rm act}\mathcal{A}(x)]$$

where $\mathcal{A}(x)$ is the activity-indicator functional and $\alpha_{\rm act}$ is the substrate-coupling. The structural scaling form is FORM-FORCED; both $\alpha_{\rm act}$ and $\mathcal{A}(x)$ are currently VALUE-INHERITED.

Test. Stratify existing weak-lensing surveys (KiDS, DES, HSC; upcoming LSST/Vera Rubin + Euclid) by activity proxies (star formation rate; AGN classification; dynamical relaxation indicators) at fixed mass + redshift. Compare lensing convergence amplitude across activity bins.

Sharpest falsifier (F3). Activity-independent weak-lensing convergence at deep-MOND-regime radii across all examined activity bins refutes the V5-mobility-gradient mechanism (Paper_038_6 §3.4).

ED-vs-CDM-with-feedback discriminator status. Standard CDM-with-feedback also predicts some activity-dependence of inferred halo mass (via baryonic feedback histories). The **discriminator observable** — a $|\nabla\mu|^2$-vs-feedback-proxy observable that distinguishes the substrate-V5 mechanism from feedback histories — is OPEN. Pred_ classification PROVISIONAL until this discriminator is constructed.

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§5 GW merger-lag existence prediction (Paper_052_5)

Prediction. Compact-binary mergers (BH-BH, NS-NS, BH-NS) exhibit a **dynamical delay** $\tau_{\rm lag}$ between the standard-relativistic predicted merger time and the substrate-level actual merger completion.

Substrate mechanism (Paper_052_5). Three substrate contributions to the delay:

1. **V5 finite-memory dissipation** (Paper_090): substrate cross-chain bandwidth needs to redistribute during merger; redistribution timescale is τ_{V_5}.
2. **Mobility-gradient acoustic-metric curvature** (Paper_EDSC_AcousticCurvature): merger events involve large local mobility gradients; acoustic-metric curvature on these scales modifies effective dynamics.
3. **Kernel cascade across scales** (Paper_091): merger events span multiple substrate scales (compact-object substrate-cell scale → merger-event scale → post-merger-relaxation scale).

The **existence** of the delay is form-IDENTIFIED. Specific numerical lag magnitudes are INHERITED.

Honest scope. This is *not* a refutation-grade prediction in the BTFR-slope-4 sense. The substrate-anchored floor $\tau_{V_5}^{\rm min}$ is OPEN; only existence-of-delay is predicted at refutation-grade level. Pred_ classification PROVISIONAL pending substrate-anchored τ_{V_5} floor.

Test. LIGO/Virgo/KAGRA + future Einstein Telescope / Cosmic Explorer / LISA: residual analysis of compact-binary merger waveforms against standard-relativistic predictions. Existence of systematic lag (positive sign, multiple uncorrelated events) supports the substrate-V5 mechanism.

Sharpest falsifier (F4, conditional). Once substrate-anchored $\tau_{V_5}^{\rm min}$ floor is supplied: observation of merger-lag systematically *below* the derived floor across uncorrelated systems refutes the substrate mechanism.

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§6 Quantum-to-classical mass wall at 140–250 kDa (QC-Mass-Extrapolation protocol)

Prediction. In matter-wave interferometry, the coherence fraction $V_{\text{coh}} = V_{\text{measured}}/V_{\text{env}}$ crosses a quantum-to-classical (Q-C) transition at $V_c \approx 0.304$. A two-point coherence trajectory anchored at Eibenberger 2013 (~10 kDa, $V_{\text{coh}} \approx 0.82$) and Fein 2019 (~25 kDa, $V_{\text{coh}} \approx 0.76$) extrapolates the crossing to **molecular mass** $m_c \in [140{,}000, 250{,}000]$ **amu** — 5–10× beyond current Arndt-group experimental reach.

Secondary distinguishing prediction. In the $D < 0.1$ sector approached as $V_{\text{coh}} \to V_c$, ED predicts **second-harmonic interference content at 3–6%** in fringe data residuals. Standard decoherence theory predicts pure first-harmonic decay. Measurable in residuals of sinusoidal fits to fringe data near the crossing.

Test platform. Long-baseline matter-wave interferometry — LUMI-class (Arndt group, Vienna) or successor instrument capable of $m > 100$ kDa. Candidates: - **LUMI extension** (Vienna) — natural home; designed extensible - **MAQRO / space-based concepts** — design mass reach $m \sim 10^6$ amu; >10-year horizon - **Next-generation OTIMA / Talbot-Lau variants** — under discussion in Arndt-group roadmap

Sharpest falsifier (F5). V_{coh} crossing at $m_c < 100{,}000$ amu or $m_c > 400{,}000$ amu refutes the two-point extrapolation under either linear or exponential fit.

Protocol status. Stub proposal drafted 2026-04-28; three blockers before fully-specified protocol (promotion of affine $D_P \to D_E$ mapping from CANDIDATE to FORCED; apparatus envelope $V_{\text{env}}(m)$ characterization for $m > 100$ kDa; molecular library at target mass scale). **Primary collaboration targets:** Markus Arndt (Universität Wien); Yaakov Fein (Cambridge); Stefan Gerlich (Vienna); Klaus Hornberger (Duisburg-Essen). Secondary: MAQRO consortium.

Pre-registration epistemic move. The prediction is publicly pre-registered (Zenodo DOI + this paper) before any experiment in the target range is published. Stub-now-execute-later is the appropriate posture for an outsider research program.

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§7 FRAP mobility scaling in concentrated BSA (UDM law protocol)

Prediction. Fluorescence Recovery After Photobleaching (FRAP) in concentrated BSA-FITC at 200–350 mg/mL follows the ED Universal Degenerate-Mobility (UDM) law $M(\rho) = M_0(\rho_{\max} - \rho)^\beta$ with canonical $\beta \approx 2$. This produces the 2D porous-medium equation with $m = \beta + 1 = 3$, giving front-radius exponent:

$$\alpha_R = \frac{1}{d(m-1)+2} = \frac{1}{6} \approx 0.167.$$

Predicted observable: $R(t) \sim t^{1/6}$ with **sharp** bleach boundary (compact support, PME signature).

Null prediction (Fickian). $R(t) \sim t^{1/2}$ with Gaussian-blurred boundary.

Test platform. Standard confocal FRAP on concentrated BSA-FITC at 200, 250, 300, 350 mg/mL ($\phi = 0.15, 0.18, 0.22, 0.25$). 5 bleaches per concentration, 20 total acquisitions.

Sharpest falsifier (F6). Observed $\alpha_R \geq 0.35$ or Fickian-front winning at all concentrations refutes the UDM mobility-channel mechanism. PASS criterion: $\alpha_R \in [0.125, 0.170]$ at $\geq 3/4$ concentrations + PME-front beats Fickian via AIC.

Protocol status. Protocol submitted to Creative Proteomics CIBR on 2026-04-17; decision window 2026-04-24 to 2026-05-01. **Budget:** \$500–$1500 for session; 6\breve{}8 weeks calendar time to publishable result. * * Independent second confirmation * * of the UDM 10 - material result$(R^2 > 0.986\$) via a different observable (front propagation rather than diffusion-coefficient regression). The peer-reviewed Universal Mobility Law paper (May 2026) provides the prior empirical anchor; see §11 below.

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§8 AFM thin-film polymer dewetting cross-scale test (AFM-Dewetting-ED-SC protocol)

Prediction. ED-SC 2.0 architectural invariance: the motif-conditioned median of the field-space Hessian eigenvalue-ratio distribution satisfies

$$\mathrm{med}(\mathcal{R}_{\mathrm{motif}}) = r^* \approx -1.30 \in [-1.50, -1.10]$$

on 2D real-space scalar density fields. Reference system: Scenario D at $(n^*, \sigma^*) = (2.7, 0.0556)$ gives $\mathrm{med}(\mathcal{R}_{\mathrm{motif}}) = -1.304$.

Test platform. AFM measurement of height field $h(x, y)$ of spin-coated polystyrene on silicon, annealed into the spinodal pre-rupture regime, tapping-mode AFM at $\Delta x \leq L_{\mathrm{coh}}/8$ resolution.

Sharpest falsifier (F7). Median $\mathcal{R}_{\mathrm{motif}}$ outside $[-1.80, -0.90]$ refutes the ED-SC 2.0 architectural invariance claim. PASS criterion: median in $[-1.50, -1.10]$ within $\epsilon_{\mathrm{med}} = 0.20$ of reference r^*.

Protocol status. Protocol complete and session-ready. **Budget:** ~\$500 AFM facility + 2 weeks analysis; 4–6 weeks calendar time to publishable result. **Quoted at \$15K** for full slide-prep-to-data-collection turnkey (commercial). Three execution routes available: (1) reanalysis of existing 2023 *Nat Comm Phys* AFM data (5-min check), (2) collaboration with Jacobs group at Saarland, (3) book a 1–2 day session at any academic core facility with PS-on-Si dewetting samples.

Why this test matters. ED-SC 2.0 is the cross-scale architectural invariance claim — independent of any specific PDE channel. AFM dewetting tests *motif-level* mobility-curvature geometry, not a single channel. PASS validates ED-SC 2.0 at substrate-architectural level; FAIL refutes the architectural invariance claim.

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§9 Acoustic / BEC analogue-gravity extremal-horizon tests

Prediction. ED acoustic metric + Canon P4 mobility-collapse → interior-maximum horizons are **extremal** ($\kappa = 0$, no Hawking-analogue phonon pair emission). Three sibling experiments test this in different analogue-gravity platforms:

§9.1 ED-Acoustic-EIT-Extremal (top priority)

Observable. Thermal-emission-rate ratio R_E/R_M between extremal (E) and monotonic (M) control-field configurations in the same EIT atomic cell. **Prediction:** $R_E/R_M < 0.1$ (PASS); > 0.5 (FAIL).

Platform. Spatially-shaped EIT control field on warm Rb vapour or cold Rb ensemble, toggled by SLM between monotonic and extremal $v_g(x)$ profiles. **Methodological strength:** within-apparatus differential — same cell, same detection chain, eliminates systematic drift.

Status. Protocol drafted 2026-04-22. **Primary collaboration targets:** Lukin (Harvard); Hau (Harvard); Halfmann (Darmstadt); Lvovsky (Oxford); Giacobino/Bramati (LKB Paris).

§9.2 ED-Acoustic-BEC-Extremal (highest-profile follow-up)

Observable. Density-density correlation function $G(x, x')$ peak amplitude at Hawking-pair null-ray locus; phonon momentum-distribution thermal tail. **Prediction:** correlation peaks and thermal tail suppressed in extremal configuration relative to monotonic (Steinhauer-standard) configuration.

Platform. Elongated 1D Rb-87 BEC with custom trap + flow engineering producing smooth second-order zero of $v^2(x) - c_s^2(x)$ at interior point.

Status. Stub protocol; blocked on moving-background ED acoustic-metric theory extension + trap-flow co-design + within-apparatus differential protocol. **Primary collaboration targets:** Steinhauer (Technion); Westbrook (Institut d'Optique); Cornell/Jin (JILA); Ferrari (Trento).

§9.3 ED-Acoustic-OpticalPulse

Observable. Pulse-peak (extremal $\kappa = 0$) vs pulse-flank (monotonic $\kappa > 0$) differential pair-emission within a single optical pulse propagation apparatus.

Status. Stub protocol; second priority in the acoustic-analogue program.

Sharpest cross-platform falsifier (F8). Consistent thermal-emission detection in extremal-configuration analogue horizons across multiple platforms (EIT + BEC + optical-pulse) refutes the ED P4 mobility-collapse → extremal-horizon mechanism.

Honesty statement. $\kappa = 0$ at smooth maximum is a *kinematic* property of any acoustic metric, not ED-specific in isolation. These tests discriminate *between configurations within one acoustic-metric framework*, which ED adopts. PASS supports ED's P4-is-extremal reading consistent with the kinematic prediction; FAIL ruptures either the acoustic-metric identification or ED's extremal-horizon mapping.

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§10 ED-09.5 Participation-Envelope prediction

Prediction. ED-09.5 is the corpus's **single most distinctive falsifiable prediction**: the canonical ED PDE's damping-discriminant bifurcation at $D_{\text{crit}} = 0.5$ (Canon P6) predicts that systems in the oscillatory regime ($D < 0.1$) carry a **slow envelope modulation** of their dynamics at frequency:

$$\omega_v = 2\pi N_{\text{osc}} \gamma_{\text{dec}} \approx 8\gamma_{\text{dec}} \quad \text{(single-mode, } N_{\text{osc}} = 8\text{)}.$$

Quantum-side signatures include $Q_v \in [4, 9]$, third-harmonic content 3–6%, triad coupling $C \in [0.01, 0.05]$.

§10.1 Two parallel test tracks (same prediction, regimes separated by 10+ OOM in γ_{dec})

Track A — Cavity-coupled optomechanics reanalysis. Target datasets: Delic 2020 (levitated SiO) and Magrini 2021 (real-time feedback) at $D \sim 10^{-9}$, predicted $\omega_v/2\pi \approx 16$ mHz, envelope period ~63 s, required record length ≥ 200 s. **Observable:** slow envelope on raw heterodyne $x(t)$ after demodulation at ω_m. **Status:** blocked on raw data access; Aspelmeyer email v3 drafted, not yet sent.

Track B — FRAP-side Lomb-Scargle on public recovery curves. Target datasets: concentrated-FRAP datasets with sampling rate ≥ 1 kHz (Mueller et al. 2008, Kang et al. 2012, Wachsmuth et al. 2003). Predicted $\omega_v/2\pi \in [80, 800]$ Hz. **Observable:** Lomb-Scargle periodogram of residual $r(t) = I(t) - I_{\text{model}}(t)$. **Status:** self-contained, executable within ~1 week at 0 — **fastest path to a new ED empirical result in the entire program**.

Sharpest falsifier (F9). Full PASS on both tracks (mHz envelope in optomechanics + 100-Hz-scale envelope in FRAP residuals) would be the strongest possible confirmation of ED-09.5. Independent FAIL on Track B (no envelope detected at predicted frequency band in any of 3+ uncorrelated FRAP datasets) refutes the ED-09.5 bifurcation mechanism at the cell-biology regime.

Why this is the most distinctive prediction. ED-09.5 is regime-agnostic — depends only on the system's decoherence rate γ_{dec}. The two tracks test the same bifurcation in regimes separated by **10+ orders of magnitude** in γ_{dec}. Substantively distinct from any phenomenological-fit prediction.

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§11 Passed test — Universal Degenerate-Mobility (UDM) Law (peer-reviewed, May 2026)

Test passed. Universal $D(c) = D_0(1-c/c_{\max})^\beta$ functional form derived from ED PDE substrate framework. Tested against **10 materials** spanning soft-matter / biological systems. **Result:** $R^2 > 0.986$, peer-reviewed publication ("Universal Degenerate-Mobility Scaling in Crowded Soft Matter"), May 2026.

Significance. This is ED's only **peer-reviewed external-validation-already-achieved** result. The universal $D(c)$ functional form is derived from the ED mobility-channel (Canon P4) prior to any data fitting. The peer-reviewed match validates the substrate-level identification at observational accuracy.

Status. PASSED. Empirical confirmation supports the ED mobility-channel framework. Follow-on prediction (§7 above): FRAP-BSA front-radius exponent $\alpha_R = 1/6$ provides independent second confirmation via a different observable.

Sharpest falsifier (F10, hypothetical). A material in the UDM-application class with $R^2 < 0.8$ against the universal $D(c) = D_0(1 - c/c_{\max})^\beta$ functional form would refute the substrate-level universality claim. Currently consistent across all 10 examined materials.

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§12 Passed test — GW170817 speed equality with c

Test passed. GW170817 (binary-neutron-star merger, August 2017): multi-messenger constraint $|c_{\text{GW}} - c|/c < 10^{-15}$. ED's prediction (Paper_ED_GW_00 §3.8): GW propagation speed =

substrate-c, the same c that bounds electromagnetic propagation per Paper_012 (P-RB-1). ED structurally inherits GW speed $= c$; falsification would have required $|c_{\mathrm{GW}} - c|/c \gtrsim 10^{-15}$.

Significance. GW170817 ruled out a large class of modified-gravity theories that predict $c_{\mathrm{GW}} \neq c$ (Tensor-Vector-Scalar theories, certain massive-graviton models, certain dark-energy theories). ED's substrate-graph reading — GW = propagating saddle-Hessian reconfiguration carried by V1 retarded kernel at substrate-c (Paper_ED_GW_00 §3.3) — is structurally consistent with the observation. **ED was not falsified by GW170817; many alternative-gravity programs were.**

Sharpest falsifier (F11, hypothetical). Future GW + electromagnetic multi-messenger event with $|c_{\mathrm{GW}} - c|/c \gtrsim 10^{-15}$ would refute Paper_ED_GW_00's substrate-c identification of GW propagation.

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§13 Postdiction — -smallness 120-OOM reframing (Paper_038_5; M3 retroactive)

Postdiction. The famous cosmological-constant problem ($\rho_\Lambda/\rho_P \sim 10^{-120}$) is **reframed**, not paradoxical, at substrate-graph level.

Substrate-graph reframing (Paper_038_5 + Paper_ED_Cos_05).

1. ED's **discrete substrate ontology does not admit infinite QFT mode-tower zero-point summation**. The naïve QFT estimate $\rho_{\mathrm{vac}}^{\mathrm{naive}} \sim M_P^4$ is *structurally absent* substrate-side — there is no infinite mode tower to sum.
2. Substrate-graph closure of ρ_Λ reduces to Friedmann inheritance: $\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$ with M_P^2 via Paper_027's dimensional rearrangement and H_0 via Route A4 substrate-graph chain (Paper_ED_Cos_05).
3. Route A4 closure (2026-05-17, see Update_Paper_ED_Cos_05_M3_Unconditional + Update_Paper_038_5_M3): $H_0 = (8\pi/3)^{1/2}\ell_P c^{1/2}\Theta_{\mathrm{ED}}^{1/2}$ with $\Theta_{\mathrm{ED}}^{\mathrm{Planck-units}} \approx 10^{-122}$ substrate-parameter-INHERITED at primitive level (analog of Paper_027's $\ell_P, \hbar, c$ inheritance).

Result. The 120-OOM "mismatch" becomes a **substrate-parameter-content value** $\Theta_{\mathrm{ED}}^{\mathrm{Planck-units}} \approx 10^{-122}$, parallel to the longstanding question "what determines $\ell_P, \hbar, c$ values?" at primitive substrate-constant level. The famous-problem framing is dissolved; the substrate-parameter-content question remains as tightening work (RA-OPEN-4a; not load-bearing for M3).

Sharpest falsifier (F12). Independent observational evidence that $\rho_\Lambda \neq (3/8\pi)H_0^2 M_P^2$ at substrate-parameter-INHERITED level would refute either Route A's substrate-graph chain or the Friedmann inheritance step. Currently consistent with Planck CMB + Type Ia supernovae + BAO at observed H_0.

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§14 Cross-scale falsifiers — SCBU synchronized-covariation prediction

Prediction. The Substrate-Cosmology Boundary Unification (SCBU; Paper_SCBU + ED-SC 4.x arc) framework predicts **synchronized co-variation of six projections** of the substrate-cosmology boundary at $R_H = c/H_0$ under Hubble-tension-class shifts of H_0:

- $\xi_{\mathrm{canonical}}$ (substrate-scale anchor; Paper_ED_SC_4.2 / Paper_096)
- $a_0 = cH_0/(2\pi)$ (MOND transition acceleration; Paper_029)

- $R_H = c/H_0$ (cosmic decoupling surface; Paper_028)
- $\mathcal{M}_{\rm crit}$ (Q-Compute platform-coherence wall; Paper_060)
- $Q \approx 3.5$ (NS-Q canonical hydrodynamic; Paper_080)
- BH horizon r_H (parameterized by local mass; Paper_ED_SC_4.1)

Post-Route-A-closure status (2026-05-17). With Route A now closed at substrate-parameter-INHERITED level (Update_ED_SC_4x_Arc_M3), per-projection scaling exponents are derivable in principle via the Route A inheritance chain:

- $a_0 \propto H_0^1$ — structural per Paper_029
- $R_H \propto H_0^{-1}$ — structural per Paper_028
- $\xi_{\rm canonical} \propto H_0$ at leading order via Route A4
- $\mathcal{M}_{\rm crit}, Q$ — per-projection derivation tightening work (substrate-research-frontier; not load-bearing for M3)

Sharpest falsifier (F13). Empirical **asynchrony** between any two SCBU-projection observables under Hubble-tension-class H_0 shifts — e.g., a_0 tracking H_0 as predicted while $\mathcal{M}_{\rm crit}$ remains flat — refutes the SCBU-projection identification for the asynchronous quantity regardless of per-projection exponent derivation status.

Why this is structurally distinctive. SCBU is the corpus's "single substrate, six projections" claim. Hubble tension is currently a live observational debate (early-universe vs late-universe H_0 measurements disagree at $\sim 5\sigma$). Independent shifts in a_0, $\mathcal{M}_{\rm crit}$, Q — observable in galactic-rotation, matter-wave-interferometry, and soft-matter platforms respectively — would test the SCBU synchronization at empirical level.

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§15 Testability summary table

#	Prediction	Source paper	Test platform	Status	Sharpest falsifier
1	BTFR slope-4 + zero scatter	Paper_031	SPARC catalog (public)	**Testable now**	Slope $\neq 4$ or intrinsic scatter at deep-MOND-asymptote
2	$a_0 = cH_0/(2\pi)$	Paper_029	Hubble-tension-class H_0 refinement	**Testable as H_0 precision improves**	a_0 shift inconsistent with $cH_0/(2\pi)$
3	Weak-lensing activity-dependence	Paper_038_6	KiDS / DES / HSC / LSST / Euclid	**Testable now** (PROVISIONAL Pred_)	Activity-independent lensing at deep-MOND radii
4	GW merger-lag existence	Paper_052_5	LIGO/Virgo/KAGRA + ET / CE / LISA	**Testable now** (PROVISIONAL Pred_)	Merger-lag $< \tau_{V_5}^{\rm min}$ (floor OPEN)

#	Prediction	Source paper	Test platform	Status	Sharpest falsifier
5	QC mass wall 140–250 kDa	QC-Mass protocol	LUMI / OTIMA / MAQRO	**Pre-registered**; needs LUMI extension	V_{coh} crossing $m_c < 100$k or > 400k amu
6	UDM front-radius $\alpha_R = 1/6$	FRAP-BSA protocol	Confocal FRAP (CP submitted)	**Protocol active**; 6–8 wk timeline	$\alpha_R \geq 0.35$ or Fickian wins at all conc.
7	$\text{med}(\mathcal{R}_{\text{motif}}) = -1.30$	AFM protocol	Tapping-mode AFM on PS/Si	**Session-ready**; $15K quoted	Median outside $[-1.80, -0.90]$
8	Extremal-horizon zero pair emission	EIT/BEC/Optical protocols	EIT Rb cell + 1D Rb-87 BEC	**EIT protocol drafted**; collab pending	Pair-emission in extremal configuration
9	ED-09.5 envelope $\omega_v \approx 8\gamma_{\text{dec}}$	ED-09.5 protocol	Optomechanics + FRAP residuals	**Track B: $0, 1-week executable**	No envelope at predicted band
10	UDM $D(c) = D_0(1 - c/c_{\max})^\beta$	UDM peer-reviewed paper	10-material soft-matter dataset	**PASSED** ($R^2 > 0.986$)	(Hypothetical: material in class with $R^2 < 0.8$)
11	GW speed $= c$	Paper_ED_GW_C	GW170817 multi-messenger	**PASSED** ($\lvert c_{\text{GW}} - c\rvert/c < 10^{-15}$)	Future multi-messenger $\lvert c_{\text{GW}} - c\rvert/c \gtrsim 10^{-15}$
12	smallness $= \Theta_{\text{ED}}^{\text{Planck-units}}$	Paper_038_5 (M3 retro.)	Planck + SNe Ia + BAO	**CONSISTENT** with $\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$	ρ_Λ inconsistent with Friedmann form
13	SCBU synchronized covariation	ED-SC 4.x arc	Multi-axis observational stratification	**Testable as H_0 measurements refine**	Asynchrony between any two projections

Three predictions are testable now at low cost (#1 SPARC reanalysis; #9 FRAP-residual Lomb-Scargle reanalysis; #10/#11/#12 already-passed). **Two are protocol-ready** with budgets quoted (#6 FRAP-BSA at ~$1500; #7 AFM at ~$500–15K). **Five await collaboration with experimental groups** (#5 LUMI/Arndt; #8 EIT/BEC analogue-gravity; #9 Track A Aspelmeyer optomechanics). **Three are observational-pipeline analyses** (#3 weak-lensing survey reanalysis; #4 GW residual analysis; #13 cross-axis Hubble-tension stratification).

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§16 What this paper does and does NOT claim

This paper does **not** claim: - That ED replaces the Standard Model, CDM, or general relativity. ED supplements standard physics with a substrate-level account; standard-physics predictions remain excellent within their regimes. - That every prediction listed is refutation-grade (in the BTFR-slope-4 sense). Where prediction status is PROVISIONAL Pred_ (e.g., Paper_038_6 weak-lensing; Paper_052_5 merger-lag), this is named explicitly per Paper_095's verdict grammar. - That substrate primitives are derived from a deeper layer. The 13 primitives (Paper_087) are posited; the empirical case rests on cross-domain reach. - Sole-theory status. Many predictions listed (BTFR slope-4; a_0; ρ_Λ) are consistent with MOND-as-phenomenology or other approaches at observational level; ED's claim is that it *derives* these from a small set of substrate primitives whereas other approaches fit or postulate them.

This paper does claim: - ED's substrate-level identification produces a family of cross-domain predictions traceable to the substrate-graph identification $a_0 = cH_0/(2\pi)$ (the wedge) and the substrate-cosmology boundary $R_H = c/H_0$ (the organizing claim). - The form-FORCED structural content of each listed prediction is derived from substrate primitives per Paper_095 discipline. - Each prediction is paired with a sharpest falsifier that is operationally explicit and observationally accessible. - The corpus's substrate-research-pattern progression (Load-Bearing Program → Q1Q2 + Chain-Class Identification → Route A audit cascade) has delivered the Cosmology Arc M3 Trinity (Cos_01 + Cos_05 + Paper_038_5 all at M3) + ED-SC 4.x arc-wide M3 upgrade as of 2026-05-17.

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§17 References

Source corpus papers:

- Paper_027 — Newton's G from substrate constants
- Paper_028 — Cosmic decoupling surface $R_H = c/H_0$
- Paper_029 — Transition acceleration $a_0 = cH_0/(2\pi)$
- Paper_030 — ED Combination Rule $a = \sqrt{a_N a_0}$
- Paper_031 — BTFR slope-4 with zero intrinsic scatter
- Paper_034 — Deep-MOND asymptotic regime
- Paper_036 — MOND field equation
- Paper_038_5 — Lambda V1 Cosmological (-smallness reframing; M3 retroactive)
- Paper_038_6 — Pred_WeakLensing_Activity
- Paper_047 — Hawking spectrum $T_H = \kappa/(2\pi)$ via V5 KMS
- Paper_052_5 — Pred_MergerLag
- Paper_060 — $\mathcal{M}_{\text{crit}}$ unification
- Paper_073 — DCGT (Diffusion Coarse-Graining Theorem)
- Paper_080 — NS-Q canonical $Q \approx 3.5$
- Paper_087 — 13 primitives (position paper)
- Paper_089 — V1 retarded kernel + T18
- Paper_090 — V5 cross-chain kernel
- Paper_095 — Form-FORCED / value-INHERITED methodology
- Paper_096 — Cross-scale invariance + $\xi_{\text{canonical}}$
- Paper_SCBU — Substrate-Cosmology Boundary Unification
- Paper_ED_SC_4_1 through 4_6 — ED-SC 4.x cross-scale extension arc
- Paper_ED_Cos_01 — Inflation (M3 unconditional, 2026-05-17)
- Paper_ED_Cos_05 — Dark Energy (M3 unconditional, 2026-05-17)
- Paper_ED_GW_00 — Gravitational Waves (M3 + row 12 partial closure)
- Universal Mobility Law Evidence paper (peer-reviewed, May 2026)

Update memos (2026-05-17):

- Update_Paper_ED_Cos_05_M3_Unconditional

- Update_Paper_038_5_M3
- Update_ED_SC_4x_Arc_M3
- Update_SCBU_Synthesis_PostRouteA
- Update_ED_MEMORY_CosmologyArc_M3

Protocol references (`predictions/` folder):

- QC-Mass-Extrapolation_InProcess
- FRAP-High-BSA_InProcess
- AFM-Dewetting-ED-SC_InProcess
- ED-Acoustic-BEC-Extremal_InProcess
- ED-Acoustic-EIT-Extremal_InProcess
- ED-Acoustic-OpticalPulse_InProcess
- ED-09.5 Participation-Envelope_InProcess
- Triad-Coupling-C7_InProcess
- ED-RLC-Analogue_InProcess

External empirical anchors:

- Lelli, McGaugh, Schombert 2016 — SPARC galaxy-rotation-curve catalog
- McGaugh 2012 — BTFR slope-4 empirical measurement
- Milgrom 1983 — MOND original phenomenological framework
- LIGO/Virgo Collaboration 2017 — GW170817 multi-messenger detection
- Planck Collaboration — CMB + ΛCDM cosmological parameters
- Eibenberger et al. 2013 — Matter-wave interferometry ~10 kDa
- Fein et al. 2019 — Matter-wave interferometry ~25 kDa
- Delic et al. 2020 — Levitated optomechanics SiO
- Magrini et al. 2021 — Real-time feedback optomechanics

End Paper_ED_Predictions_Bundle.